Energy-Based Interpretation of General Relativity

A Lay Hypothesis

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1. Starting Point of GR

General Relativity describes gravitation geometrically. Matter and energy are described by the energy-momentum tensor **T_mu_nu**. The Einstein field equations read:

- **G_mu_nu = (8 * pi * G / c^4) * T_mu_nu**

Where:

- **G_mu_nu** describes the curvature of spacetime,
- **T_mu_nu** describes the local distribution of energy, momentum, pressure, and stresses.

The geometry of spacetime, in turn, determines the possible motion of energy and matter. This yields the familiar relationship:

- **T_mu_nu → g_mu_nu → Dynamics → T_mu_nu**

2. My Fundamental Modification

The hypothesis initially does not introduce a new theory of gravitation. Instead, the ontological separation between:

- matter,
- radiation,
- field energy,
- and other forms of energy

is abandoned at the fundamental level.

The core assumption is:

- **Universe = E**

The universe fundamentally consists of energy in different states, distributions, and forms of motion. In this interpretation, "matter" is not an independent building block.

- **Matter = bound energy state**

Radiation, on the other hand, is a largely free energy state.

3. What Is Bound Energy?

In this hypothesis, matter arises when several photons permanently combine into a shared, stable structure. The photons do not simply lie side by side, but can overlap significantly and jointly form a compact, slightly irregular shape—conceived in simplified terms as a small, dynamic sphere.

Within this structure, the photons move and change together around a shared center of energy and gravity without losing their strong spatial overlap. In this way, a quark or an electron, for example, could be understood as such a bound photonic structure.

Which shape such a structure takes and whether it remains stable likely depends on several properties of the participating photons—particularly their energy, spin, polarization, and mutual interactions. The properties of the photons do not necessarily have to be identical. Certain combinations or even opposing orientations could be crucial for stability.

Gravity plays an important role here, as the shared energy distribution itself could contribute back to the binding of the structure.

It is experimentally and theoretically known that photons can form bound collective states under suitable conditions. The hypothesis asks the further question of whether such photonic organizational forms—under the extreme conditions of the early universe (before, during, or after the Big Bang) and taking gravity into account—could form a fundamental basis of stable matter.

Possible Testing Approach: Matter-Antimatter Annihilation

A possible starting point for investigating this hypothesis is the annihilation of matter and antimatter. In this process, for example, an electron and a positron can react with each other and convert into two gamma photons. The rest energy of the original particles is recovered in the energy of the resulting photons, provided the original particles were at rest before the reaction and no additional energy was introduced.

Within this framework of the hypothesis, one could investigate whether this process could be described as a transition of bound photonic structures into free photons. Accordingly, the electron and positron would be different bound energy states whose structures are not preserved during annihilation. The released energy would subsequently appear in the form of new photons.

This interpretation is purely hypothetical so far. The observed gamma photons do not have to be the same photons from which the original particles would have been constructed according to this hypothesis. Standard physics describes annihilation as a quantum mechanical transition between particle states rather than a proven decomposition of bound photons. However, the decomposition into their original photons would, at first glance, be more probable within this hypothesis.

Therefore, this process does not provide proof for the hypothesis. However, it represents a concrete process that could be used to test whether a model of bound photonic structures can consistently explain the properties of electrons and positrons as well as the observed energy and momentum ratios during their annihilation.

Further Hypothesis: Individual Photons as the Basis of Matter

An alternative possibility within this hypothesis would be that elementary particles do not consist of several interconnected photons, but rather correspond to a single photon whose state of motion has fundamentally changed under certain physical conditions.

While a free photon propagates rectilinearly in a vacuum at the speed of light (c), a special state could emerge within this speculative concept in which its direction of motion shifts, for example, into a rotating, closed, or otherwise organized dynamic. The energy of the photon would not be lost in the process, but would remain confined in a bound configuration. Such a structure could potentially produce properties that are observed in elementary particles as rest mass, spin, or electric charge.

In this context, it would need to be investigated in particular whether and under what conditions the spin of a particle—such as 1/2 for electrons and quarks—as well as its electric charge could be derived from such a configuration. For quarks, it would additionally have to be explained how color charge and the associated strong interaction arise. Likewise, it would need to be examined whether such a structure can be stable and compatible with known conservation laws and experimental observations.

Extreme physical conditions in the very early universe would come into question as a possible formation mechanism within this hypothesis. It is conceivable that before or during the early cosmic expansion, photons were in a state of extremely high energy density where their interactions and the conditions of spacetime permitted a different dynamic than the free propagation of individual photons in today's vacuum. It would have to be investigated whether, when, and through what physical mechanism a transition from free photon propagation to such a bound configuration could be possible at all.

These considerations are purely hypothetical so far. In particular, it is not known whether photons can actually transition into a stable, rotating, or closed state of motion under such conditions. Furthermore, this idea must not be confused with an ordinary mechanical rotation of a photon, since its intrinsic spin is a quantum mechanical property. The hypothesis would therefore need to develop a mathematically consistent mechanism that reproduces the observed properties of elementary particles and enables testable predictions.

Initial Rough Calculation – Order of Magnitude Only

To initially get a sense of scale, a very simple sample calculation can be performed. It is by no means intended to prove that a quark consists of photons, but merely to show that theoretically, a large number of photons would not necessarily be required.

As an example, the mass of an up quark of approximately $m_u \approx 2.16$ MeV / c^2 is used. The corresponding rest energy is thus approximately $E_u \approx 2.16$ MeV.

If we now hypothetically take photons each with $E_{gamma} \approx 0.72$ MeV, we already get:

- **3 × 0.72 MeV = 2.16 MeV**

Purely in terms of energy, three photons would already reach the scale of the assumed rest energy of an up quark.

This is explicitly only an initial rough calculation. In the hypothesis, the three photons would need to form a collective bound structure and be arranged such that their total energy and total momentum yield the properties of the assumed particle. Spin, polarization, spatial overlap, interactions, and potential gravitational binding are not yet accounted for in this simple calculation.

The calculation therefore merely shows: If the hypothesis were fundamentally possible, such an elementary structure would not necessarily have to consist of a very large number of photons. A few

sufficiently energetic photons could, purely energetically, reach the order of magnitude of a quark's rest energy.

This is not proof of a photonic structure of quarks, but merely a first quantitative illustration of the order of magnitude.

4. The Energy-Momentum Tensor Remains Intact

The crucial point is: We do not replace **T_mu_nu**. On the contrary: It remains the central mathematical tool for describing energy and momentum distribution.

- **T_mu_nu = local energy, momentum, and stress configuration**

It continues to contain, in particular:

- energy density,
- momentum density,
- energy flux,
- pressure,
- stresses.

Thus, no new "matter equation" needs to be introduced for matter. A matter system continues to be described by its corresponding **T_mu_nu**.

Our modification consists of stating that matter is ontologically not an additional basic component. It is a bound, organized state of the underlying energy—in this hypothesis, presumably an organized structure of photons.

The energy and momentum distribution of this bound structure continues to be described by the energy-momentum tensor.

5. The Einstein Field Equation Remains Unchanged

Consequently:

- **G_mu_nu = (8 * pi * G / c^4) * T_mu_nu**

remains fully intact. The change on the right-hand side is not a new formula, but a change in its interpretation.

- **Standard interpretation (simplified):** Matter / Energy → T_mu_nu
- **Our interpretation:** Energy → T_mu_nu

Matter is merely a special state of this energy.

6. Energy and Geometry

Within the hypothesis, the following chain emerges:

- **E → T_mu_nu → G_mu_nu → g_mu_nu**

Where **G_mu_nu** describes spacetime curvature, while **g_mu_nu** describes the underlying spacetime geometry (metric).

Energy thus determines spacetime geometry via its energy-momentum state. Conversely, geometry determines the possible development of energy and its states of motion.

Thus, the mutual feedback loop of GR remains intact:

- **T_mu_nu ↔ g_mu_nu**

7. The Universe as a Single Total System

The hypothesis further assumes that there is no external space and no external medium into which the universe expands. Instead, the universe is understood as a single connected physical total system.

The initial state is assumed to be an extremely concentrated and high-energy sphere. This became unstable and expanded explosively. In this view, the universe would be the expanding and cooling structure of this original explosion.

The interior space created in the process is understood within the hypothesis either as an (electromagnetic) field or simply as free space. Its interaction with the energy contained within it is to be investigated within the framework of the Einstein-Maxwell equations.

Put simply, matter and other bound structures would be the cooled and organized components of this original high-energy evolution.

To be investigated: Does the Hubble constant emerge "automatically" in this system?

8. What Actually Changes in GR?

This can be stated very concisely:

Unchanged:

- The field equation **G_mu_nu = (8 * pi * G / c^4) * T_mu_nu** remains intact.
- The geometric description of gravity, covariant energy-momentum conservation, and geodesic motion are also initially retained.

Changed Ontology:

- Instead of viewing matter, radiation, and fields as distinct fundamental physical contents, the hypothesis states: Everything is an expression of energy.
- Through the properties and interactions of individual photons—such as polarization, gravity, and further properties yet to be specified—stable structures could form from which elementary particles like quarks and other forms of matter might emerge.
- Matter is understood as a stable or long-lived, organized state of this energy or these photonic structures.
- The various physical states are described regarding their gravitationally relevant properties by their respective energy-momentum tensor **T_mu_nu**.

9. How Does Special Relativity Fit In Here?

This hypothesis does not intend to modify the mathematical foundations of Special Relativity, but rather investigates whether its known results can be explained through an energy-based interpretation. If matter represents a bound, dynamic energy state and elementary particles consist of several interconnected photons, their internal motion could form the basis of physical processes and aging.

From this arises the speculative conjecture that relativistic time dilation is related to this internal photon movement: with increasing speed of the overall system, the internal dynamics could slow down until, in the hypothetical limiting case of movement at the speed of light, it would come to a complete standstill. However, a material system cannot reach this limiting case.

The mathematical relationships of SR are intended to remain unchanged; rather, it would need to be investigated whether its predictions can be derived from the assumed dynamics of bound photonic structures. Whether such a mechanism is physically possible and can quantitatively explain the observed time dilation remains an open research question.

10. Info

Although this hypothesis considers photons as possible fundamental building blocks of all matter, it is not ruled out that quarks and other elementary particles are based on other forms of energy particles or energy states. The fundamental assumption "Universe = E" remains unaffected by this.

The Role of Space and a Universal Field

Although the universe is described within this hypothesis as an electromagnetic field, it remains open whether such a field spanning all of space is indeed a necessary prerequisite for the described interactions. The considerations regarding gravity raise the further question of whether the observed effects might already be explainable purely through energy distribution and interactions between energy states, with matter understood as bound energy. In this case, space itself would not necessarily have to be conceived as an independent physical field.

A further possibility within this hypothesis would be that photons are not merely to be understood as excitations of an already existing electromagnetic field, but through their existence and their physical properties produce electromagnetic field phenomena themselves. In this interpretation, a field value of zero in a vacuum could indicate that no corresponding field phenomenon is present there, rather than an everywhere-present field merely existing in an unexcited state. Whether such an interpretation is physically possible and can correctly describe the known properties and interactions of photons remains open.

A universal electromagnetic field thus remains one possible interpretation, but within the current hypothesis, it is not a proven necessity. Whether interactions can emerge entirely from the properties and distribution of energy, whether electromagnetic field phenomena arise from the photons themselves, or whether a field permeating space is required remains an open question requiring further physical investigation.

11. Speculative Additional Hypothesis on the Origin

The hypothesis so far primarily describes the physical evolution of our universe. A further, significantly more speculative assumption concerns its origin.

It is possible that our universe itself emerged from the collapse of an earlier gravitational system. Following the formation of an event horizon, its internal evolution could have decoupled completely from the original universe. The collapsed system would then not simply be an object in a shared external space, but could represent the starting point of its own universe.

In this model, an extreme concentration of energy—for example, through energy released or redistributed during gravitational contraction, comparable to the physical processes in the contraction of compact astrophysical objects—would ultimately lead to an unstable state from which the expansion of our universe emerged.